

Insect Farmers: Macrotermitinae and *Termitomyces*

Physical Item Description



The piece is a representation of underground fungal comb structures curated by Macrotermitinae termites and the fruiting bodies of *Termitomyces* above ground. Materials (plywood, cardboard, scrap foam etc) were sourced from the discarded model building components and leftover raw materials of NUS Department of Architecture (DoA) projects to repurpose them for the current assignment in a conscious effort to limit further waste. Fungal structure based on the figures seen in Darlington (2021) and Durringer et al. (2006).

Write Up

Termitomyces are a genus of basidiomycete fungi which act as the primary food source for colonies of fungus-growing subfamily of termites known as Macrotermitinae (Wisselink et al., 2020) in an amazing display of complex farming and labour activities (Li et al., 2021), considered to be among the most sophisticated interdependent insect interactions. Macrotermitinae termites and *Termitomyces* fungi exist in an obligate relationship as neither partner has been observed in a freeliving state (Aanen, 2006). During periods of rain, the edible fruit bodies (also known as basidiocarps) of *Termitomyces* are produced on Macrotermitinae mounds as part of the sexual reproductive stage (Makonde et al., 2013). These mushrooms serve as a food source for humans – they are harvested for regular consumption (Darlington, 2021), while also being utilised as part of ethnomedicinal practices in local Asian and African communities (Tibuhwa, 2012). While people may be familiar with this interesting fungicultural mutualistic relationship, few are aware of how significant its contributions to the balance of the ecosystem are.

Tending to the Garden

The fungus is actively cultivated by the termites in subterranean nests, who provide constant replenishment of substrate and an ideal controlled microclimate for fungal development in well maintained ‘fungal gardens’ (da Costa et al., 2018). Elaborate ventilation structures are crafted in order to regulate the temperature and humidity within the termite nest to ensure that fungal growth conditions remain optimal (Korb, 2003). Once the fungal garden is large enough, intricate surface mounds are constructed to allow gaseous exchange with the external atmosphere and maintain the aerobic conditions required for *Termitomyces* to thrive (Turner, 1994). Older workers are in charge of collecting and transporting decomposing plant matter, while the young workers macerate and partly digest the dead plant material to be stored in complex three dimensional specialised structures known as ‘fungal combs’, made up of termite faecal pellets (Nobre et al., 2010) which the fungal mycelium then colonises. The termites also ensure that *Termitomyces* are not outcompeted by preventing other fungi from colonising within the nest (Rouland-Lefèvre et al., 2006).

Profits for the Farmers

Termites feed on the plant matter decomposed by the fungus as well as on vegetative structures rich in nutrients, nitrogen, and proteins termed nodules. These nodules are sprouted by the fungus and contain the asexual *Termitomyces* conidiospores (Aanen, 2006) that survive gut passage to inoculate new fungal comb substrate – effectively perpetuating this system. The fungus helps the termites to degrade the plant-derived material (e.g., wood, dry grass, and leaf litter) on which they live by producing lignocellulosic enzymes (Martin & Martin, 1978), enabling them to degrade plant cell wall polysaccharides; for example, cellulose and lignin (Veivers et al., 1991). Decomposing the chemically complex structural components of plants therefore makes it easier for the termites to ingest (Johnson et al., 1981).

Ecological Importance

The subfamily Macrotermitinae is known as one of the most efficient decomposers of plant organic matter (Aanen, 2006). In the ecosystem, this relationship plays a vital role in nutrient cycling via the digestion of cellulose rich plant biomass, such as wood and forest litter (Bastien et al., 2013). As detailed above, lignocellulosic materials are recalcitrant, thus most animals are unable to process and acquire nutrients from plant biomass (Makonde et al., 2013). In particular, in dry savannah environments where there is

low rainfall and humidity, conditions are limited in moisture and unfavourable for bacterial growth, hence decomposition by soil microbial communities struggles to take place (Bastien et al., 2013). As a result, a large proportion of plant detritus in arid ecosystems is decomposed within Macrotermitinae nests, where the *Termitomyces* fungi degrades rigid plant structures for the termites consumption and simultaneously facilitates the recycling of organic nitrogen from plant litter (Vesala et al., 2019). The simpler 'converted' debris thus becomes a more suitable substrate for further decomposition by soil microbial populations (Jouquet et al., 2005) and consequently more inorganic compounds are available for plant uptake. This fungus-termite relationship thereby functions as a considerable agent in the biological turnover of organic matter and cellulose recycling in soil environments.

Additionally, these symbionts are instrumental in carbon cycling as they process a notable amount of plant materials across multiple biomes; for example, Macrotermitinae are responsible for the degradation of approximately 90% of dead wood in semiarid Kenyan savannahs (Ferrar, 1982). The nests themselves function as carbon reservoirs; as the termites feed on plant litter containing cellulose with the help of their *Termitomyces* symbiont (Yamada et al., 2005), some of the carbon from cellulose breakdown is respired as carbon dioxide and returned to the atmosphere while the rest is stored, increasing the abundance of carbon available in the surrounding soil (Bonachela et al., 2015). Macrotermitinae-mediated carbon sequestration enriches soil, a valuable benefit for primary producers given the nutrient poor environments where such termites are typically found.

Furthermore, soil physical properties are altered and improved via termite foraging and nest building activities (Wisselink et al., 2020). As Macrotermitinae colonies burrow and create subterranean networks for fungal growth, soil particles are redistributed – it is aerated, mixed, and displaced (Bastien et al., 2013). This provides a variety of benefits to soil fertility and structural integrity, including increased soil porosity, improved soil physical stability, and better nutrient storage capacity (da Costa et al., 2018). Water infiltration is enhanced as a result, creating a more favourable substrate for vegetation to grow, where plant roots can also then penetrate the crevices created by the construction of underground termite galleries (Vesala et al., 2019). *Termitomyces*-Macrotermitinae mutualisms therefore serve a critical role in the maintenance of soil quality, especially in arid systems.

Threats

With the many ecological processes *Termitomyces*-Macrotermitinae relationships mediate, its existence is vital to a functional ecosystem. Without proper decomposition and nutrient cycling, plant nutrient availability plummets, having a domino effect on the success of flora as well as the animal communities who depend on plants as a source of food. Land use changes such as habitat disturbance and deforestation for livestock or crop farming decreases Macrotermitinae species richness and abundance through numerous factors. Suitable nesting sites become limited as land is cleared for the cultivation of agricultural crops (Rasib & Ashraf, 2014), with farmers destroying external mounds and leaving the termites unable to maintain their subterranean microclimates crucial for *Termitomyces* growth (Abe et al., 2000). Moreover, ploughing and weeding of the soil throws off the delicate balance of the carefully cultivated fungal garden by lowering humidity and increasing temperature (Wood, 1975), creating an unfavourable habitat that *Termitomyces* cannot thrive in. The application of commercial fertilisers and chemical pesticides for agricultural operations can have devastating impacts on this symbiosis – supplementing

additional synthetic sources of nutrients that are already naturally accumulated by healthy Macrotermitinae colonies would artificially alter soil chemistry that disrupts the natural nutrient cycling process undertaken by the symbionts (Darlington, 2021).

The intensive harvesting of *Termitomyces* fruit bodies also constitutes a substantial threat to its conservation. In many African countries, mushrooms belonging to this genera are considered economically important in rural communities (Koné et al., 2011). Unfortunately, *Termitomyces* grow scarcer as suitable habitats decline and harvesting pressures grow. The mushrooms are increasingly being harvested earlier and at high frequencies to meet consumer demand; however, harvesting before the fruit body can reach full maturity means that sexual spores are unable to be completely developed and released (Aanen, 2006), thereby preventing this mutualism from persisting in new colonies (Korb, 2003). Human activities, particularly agricultural expansion and overharvesting, pose a significant threat to the ecological functions and diversity of this essential fungicultural symbiosis. Management strategies can include limiting structural disturbances to Macrotermitinae nests, using organic fertilisers such as cattle manure instead of inorganic fertilisers, and improving awareness in local farmers for sustainable *Termitomyces* harvesting.

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